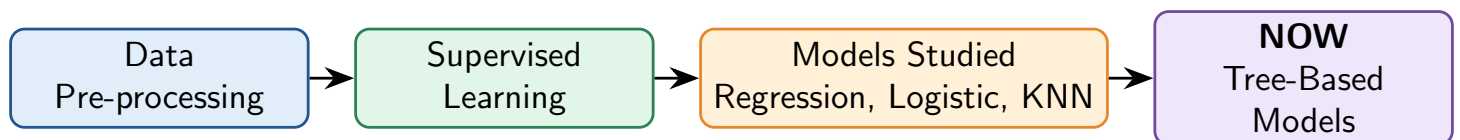


Tree-Based Models

Section 1: Introduction to Decision Trees
From basic decision-making to tree-based prediction

Machine Learning

Where Are We in Machine Learning?



Today's Question

Can we make predictions using a sequence of simple questions?

What is a Tree-Based Model?

Basic Idea

A tree-based model makes predictions by applying a sequence of:

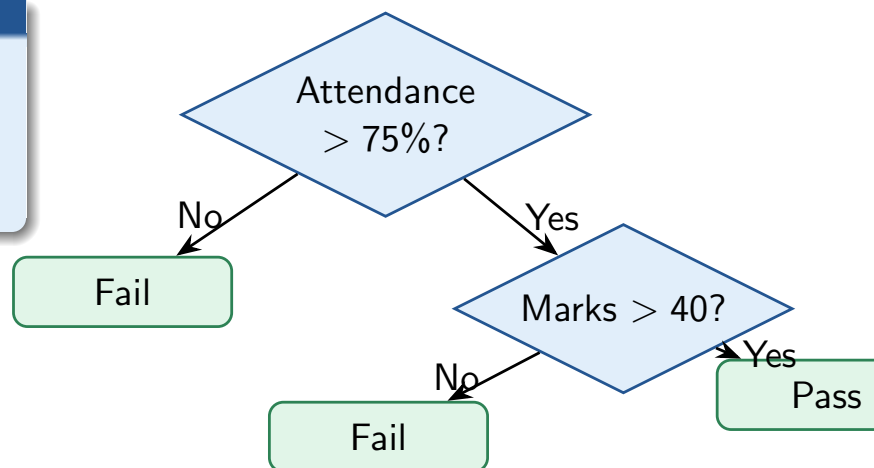
IF-THEN decisions

For example:

IF Attendance > 75%

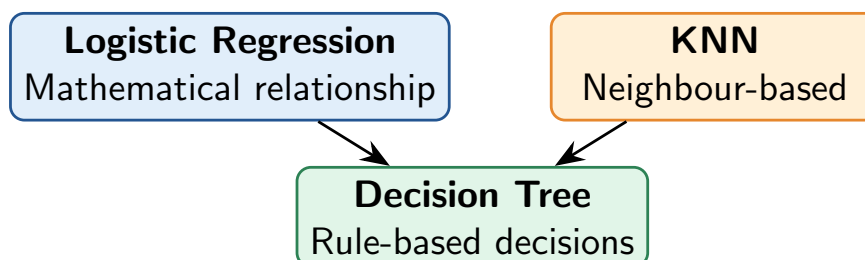
THEN check Internal Marks

ELSE predict Fail



Why Do We Need Decision Trees?

Students already know:



Main Advantage

A decision tree can represent complicated decision boundaries using **simple rules**.

- Easy to visualize and explain
- Can model nonlinear relationships
- Works naturally with decision rules

Real-World Example: Should a Student Pass?

Suppose we have:

Attendance

Internal Marks

Study Hours

Assignment

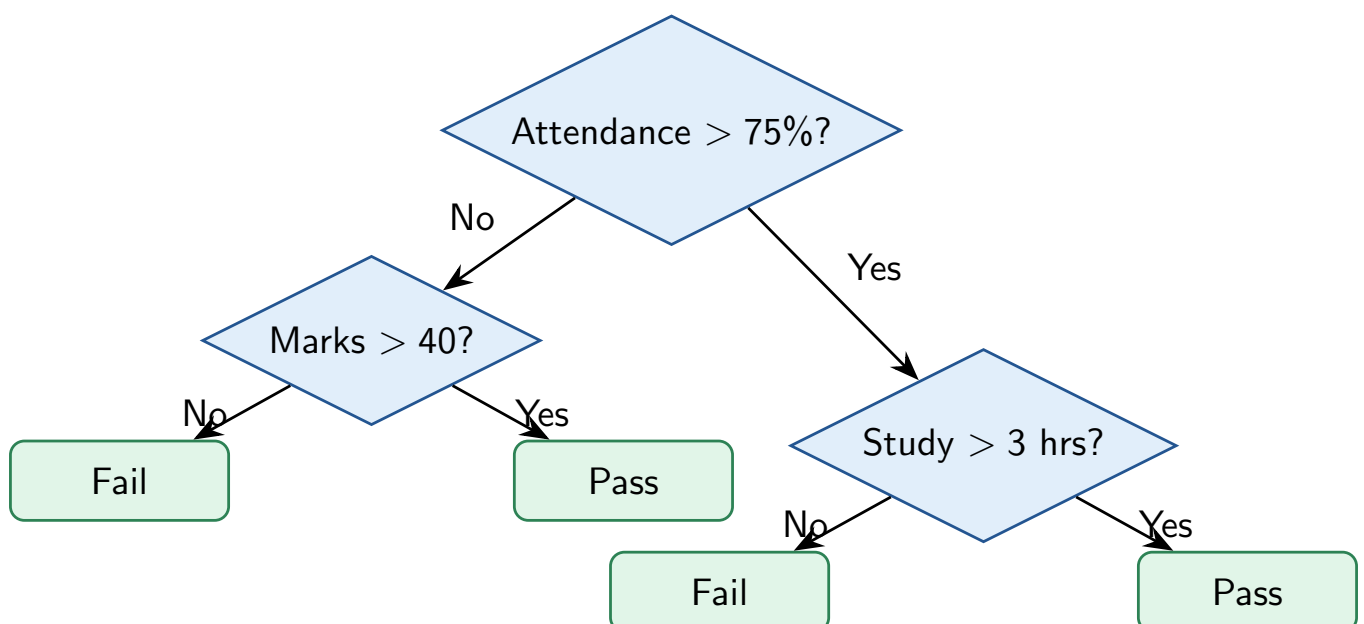
Target:

Pass / Fail

A tree tries to answer:

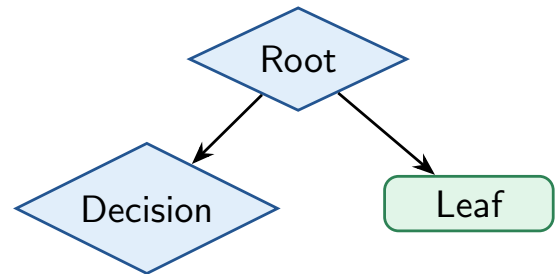
Which question should we ask first?

Decision Tree: Basic Structure



Important Terminology

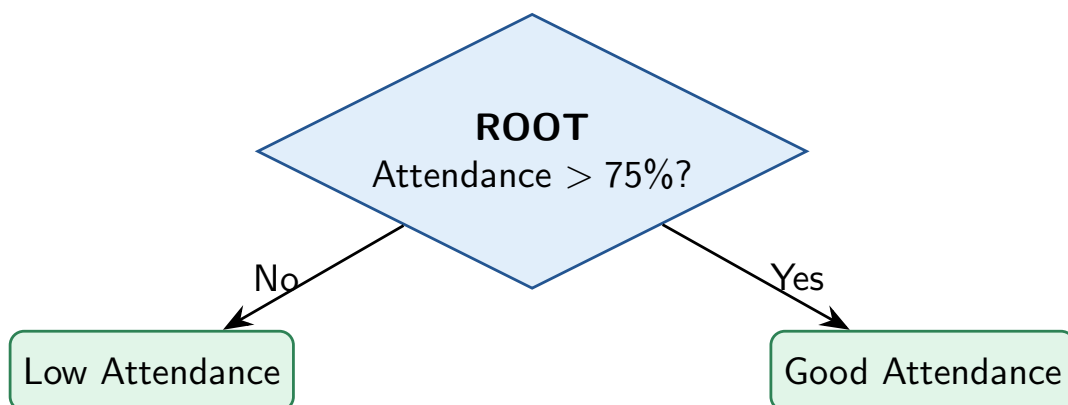
- **Root Node** — topmost node
- **Decision Node** — question used for splitting
- **Branch** — outcome of a decision
- **Leaf Node** — final prediction
- **Parent Node** — node above another node
- **Child Node** — node created after splitting
- **Depth** — number of levels in the tree



One path = sequence of decisions

Root Node: Where Does the Tree Start?

The **root** is the first decision made by the tree.

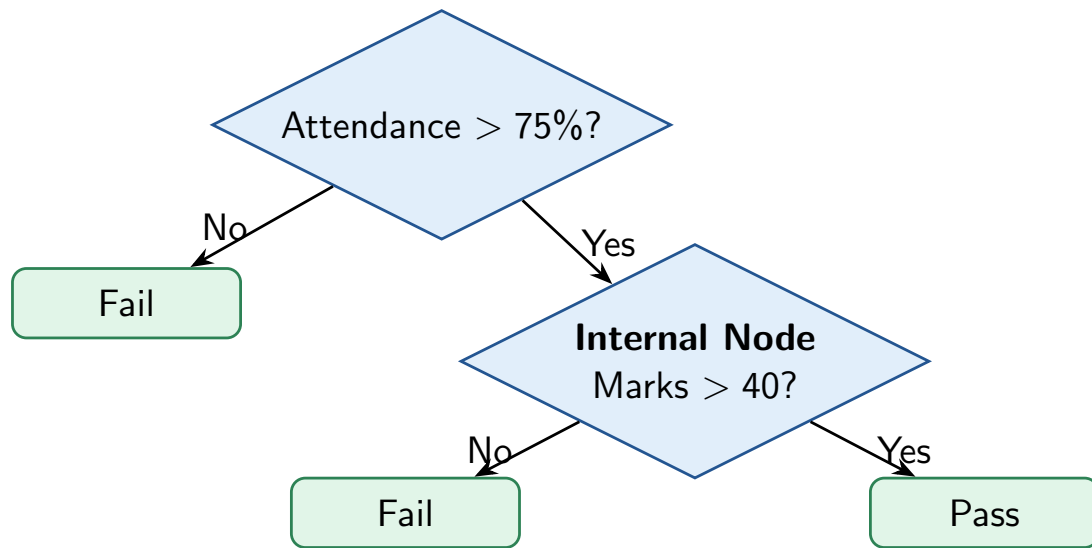


Important

The root is selected because its split gives a useful separation of the target classes.

Internal / Decision Nodes

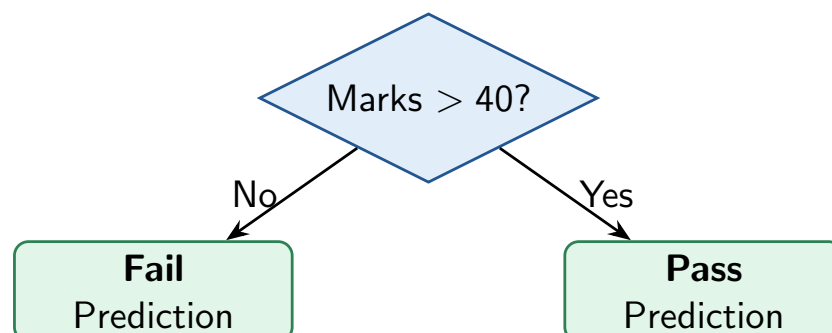
After the first split, the tree can ask more questions.



Think of an internal node as

Leaf Node: The Final Decision

A **leaf node** contains the final prediction.



- Classification tree → predicts a class
- Regression tree → predicts a numerical value

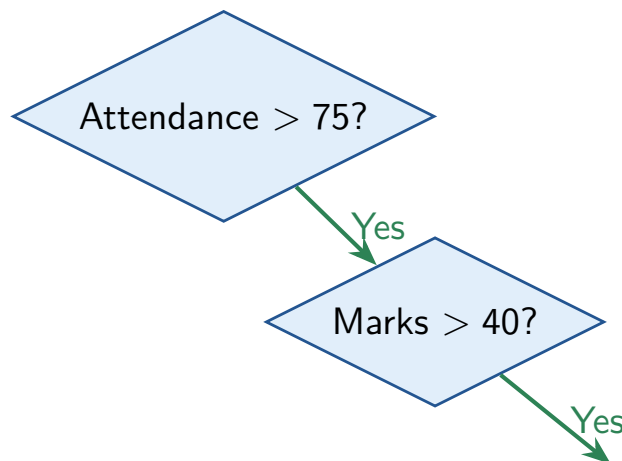
What is a Path?

A **path** is the sequence of decisions followed by one observation.

Suppose:

$Attendance = 82\%$, $Marks = 65$

Then:



Decision Tree as a Set of Rules

A tree can be converted into simple **IF–THEN** rules.

Rule 1: IF Attendance $\leq 75\%$ THEN Fail

Rule 2: IF Attendance $> 75\%$ AND Marks ≤ 40 THEN Fail

Rule 3: IF Attendance $> 75\%$ AND Marks > 40 THEN Pass

Key Idea

Every path from root to leaf represents a decision rule.

What Kind of Variables Can a Tree Use?

Recall the different types of variables from data preprocessing.

Numerical Variables

Examples:

- Age
- Income
- Height
- Marks
- Study Hours

Example split:

$Age > 30?$

Categorical Variables

Examples:

- Gender
- Blood Group
- Department
- Weather
- Yes / No

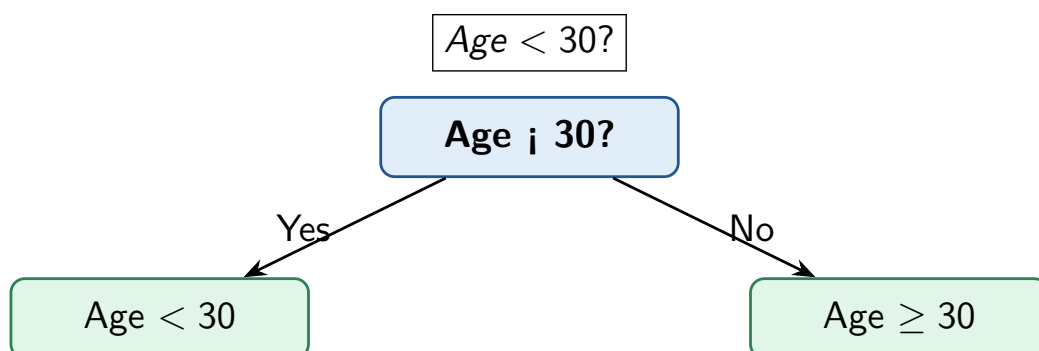
Example split:

$Weather = Sunny?$

Example: Splitting a Numerical Variable

Suppose the feature is **Age**.

Possible question:



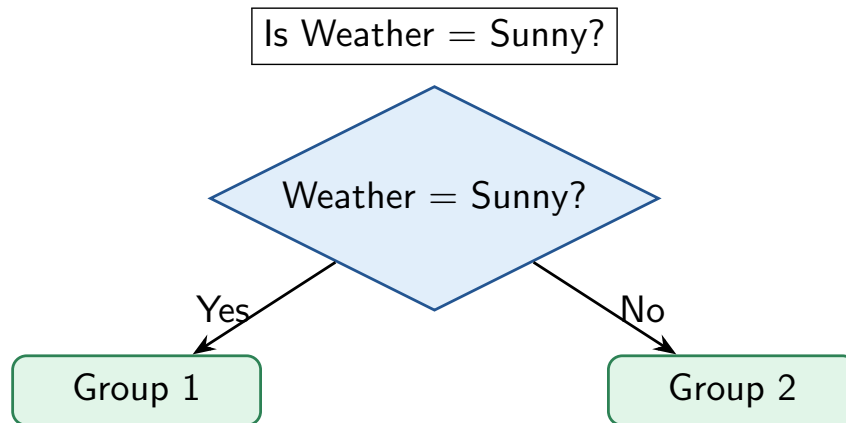
Important

A numerical variable can be divided using a **threshold**.

Example: Categorical Variable

Suppose the feature is: $\text{Weather} \in \{\text{Sunny, Cloudy, Rainy}\}$

A decision may ask:

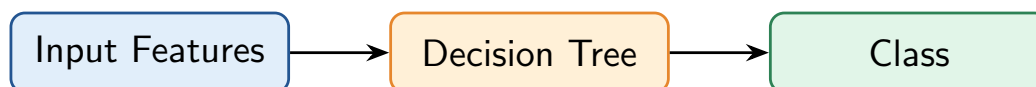


- Trees can create groups using feature values.
- The details of categorical splitting depend on the tree algorithm/software.

Decision Tree for Classification

The target variable contains **classes**.

Examples:



Examples of target classes:

Pass / Fail
Disease / No Disease
Spam / Not Spam
Benign / Malignant

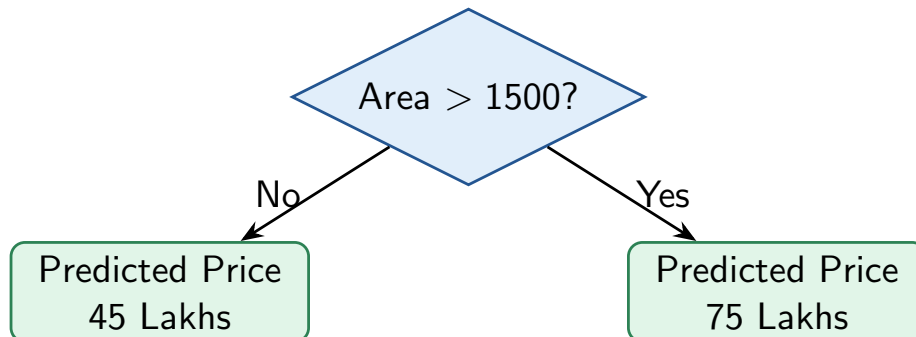
Goal

Predict the **class** of a new observation.

Decision Tree for Regression — Preview

The target variable is now **continuous**.

Example: **(House Price)** The tree may ask: **(Area > 1500 sq.ft.?)**



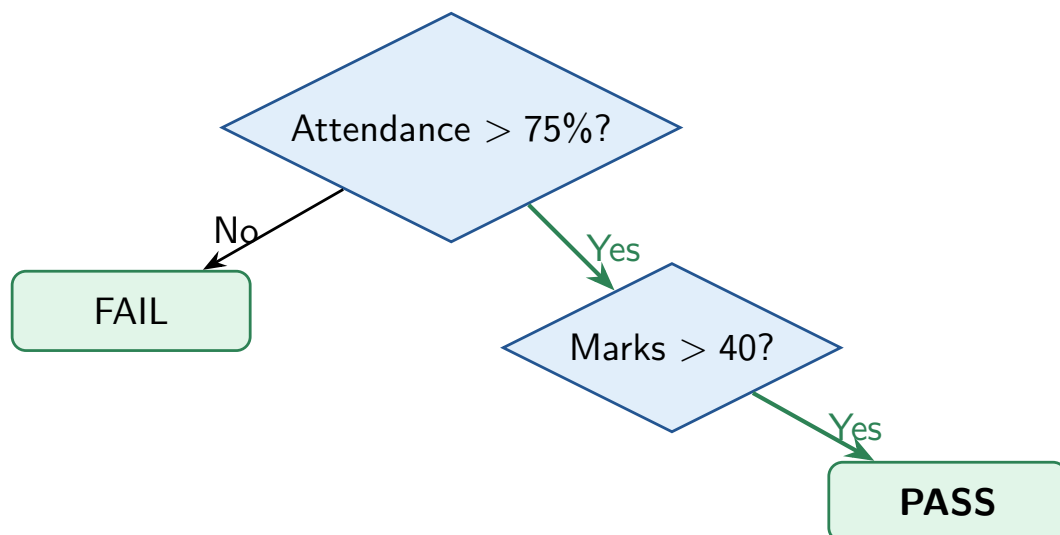
Remember

Classification → class prediction

Regression → numerical prediction

Complete Beginner Example

Suppose: Attendance=85%, Marks=62 Tree:



$85 > 75 \Rightarrow 62 > 40$ Prediction : Pass

Decision Tree vs Models Already Studied

	Logistic Regression	KNN	Decision Tree
Basic idea	Mathematical model	Neighbours	Decision rules
Prediction	Probability/class	Nearby observations	Root → Leaf
Nonlinear pattern	Limited without transformations	Can capture	Naturally captures
Interpretability	High	Moderate	Very High
Main idea	Equation	Distance	Questions

Three different ways of thinking

Logistic Regression: Equation

KNN: Neighbours

Decision Tree: Questions

Advantages of Decision Trees

Advantages

- Easy to understand
- Easy to visualize
- Easy to explain
- Captures nonlinear patterns
- Can be used for classification
- Can be used for regression

Limitations

- Can overfit
- Deep trees may become complex
- Small data changes can alter the tree
- A single tree may have limited predictive stability

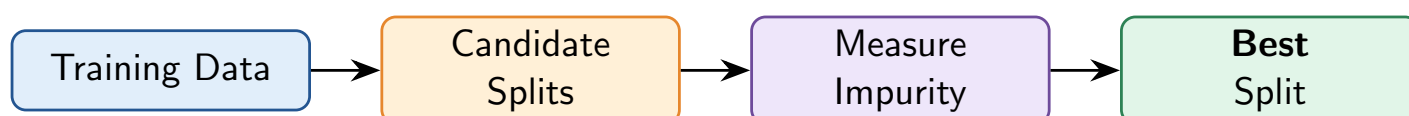
The next question:

How do we decide *which question to ask first?*

The Central Problem of Decision Trees

Everything we have seen leads to one important question:

How does the tree choose the best split?



Coming Next

Purity → Impurity → Gini → Entropy

Quick Classroom Exercise

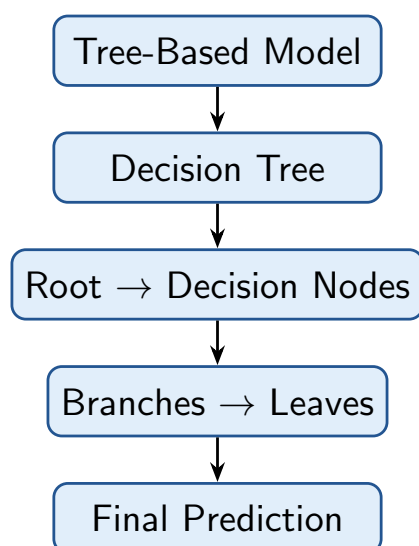
Consider the following students:

Attendance	Marks	Result
90	75	Pass
85	65	Pass
80	55	Pass
70	60	Fail
65	45	Fail
50	35	Fail

Questions

- 1 What is the target variable?
- 2 What are the features?
- 3 Is this classification or regression?
- 4 Suggest one possible first question.

Section 1 — Summary

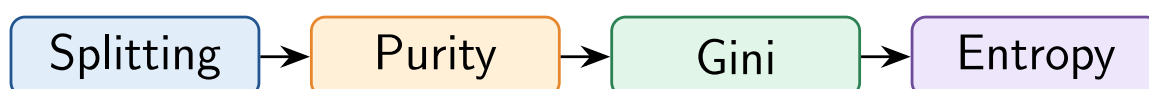


Key Takeaway

A Decision Tree makes predictions by following a sequence of **simple**

Coming Next

How Does a Tree Choose the Best Question?



The mathematics of splitting begins here.

Tree-Based Models

Section 2: Splitting, Gini and Entropy

How does a Decision Tree choose the best question?

Department of Statistics

Machine Learning

Where Did We Stop?

In Section 1, we learned:



But We Did Not Answer One Question

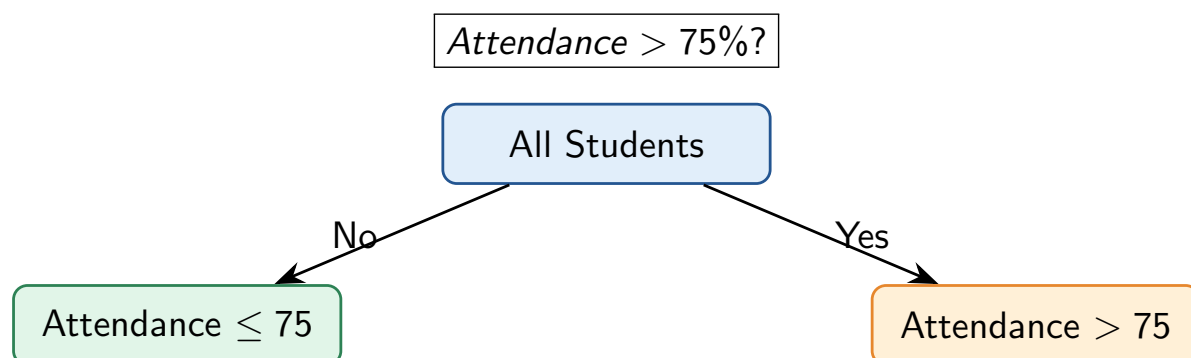
How does the tree decide which question is best?

What is Splitting?

Definition

Splitting means dividing a group of observations into smaller groups using a feature.

Example:



Why Do We Split the Data?

The main purpose is:

Create groups that are as homogeneous as possible

Suppose we want to predict: Pass or Fail

Before splitting:

{Pass, Fail, Pass, Fail, Pass, Fail}

After a good split:

{Pass, Pass, Pass} {Fail, Fail, Fail}

A Good Split

A good split separates observations into groups containing mostly one class.

Pure and Impure Nodes

Pure Node

Pass, Pass, Pass, Pass

All observations belong to the same class.

Pure

Impure Node

Pass, Fail, Pass, Fail

Different classes are mixed together.

Impure

Goal of a Decision Tree

Create child nodes that are **more pure** than the parent node.

Visualizing Purity

Pure Node



All observations:

PASS

Impure Node



Mixture of:

PASS and FAIL

Question

How can we **measure** the amount of impurity numerically?

Measures Used to Select a Split

For classification trees, two important impurity measures are:

Gini Impurity

Entropy

Both answer the same basic question:

How mixed are the classes inside a node?

Gini Impurity: Basic Idea

Gini impurity measures how mixed the classes are.

$$Gini = 1 - \sum_{k=1}^K p_k^2$$

where:

p_k = proportion of observations belonging to class k

For two classes:

$$Gini = 1 - p_1^2 - p_2^2$$

Interpretation

- Smaller Gini $\rightarrow$ purer node
- Larger Gini $\rightarrow$ more mixed node
- Gini = 0 $\rightarrow$ completely pure

Gini Example 1: Completely Pure Node

Suppose a node contains:

10 Pass, 0 Fail

Then:

$$p_{\text{Pass}} = \frac{10}{10} = 1, p_{\text{Fail}} = \frac{0}{10} = 0$$

Therefore:

$$Gini = 1 - (1)^2 - (0)^2$$

$$Gini = 0$$

Meaning

The node contains only one class, so it is perfectly pure.

Gini Example 2: Mixed Node

Suppose a node contains:

5 Pass, 5 Fail

Then:

$$p_{\text{Pass}} = \frac{5}{10} = 0.5, p_{\text{Fail}} = \frac{5}{10} = 0.5$$

Therefore:

$$Gini = 1 - (0.5)^2 - (0.5)^2 = 1 - 0.25 - 0.25$$

$$Gini = 0.5$$

Interpretation

This node is highly mixed.

Gini Example 3: Partially Pure Node

Suppose:

8 Pass, 2 Fail

Then:

$$p_{\text{Pass}} = 0.8, \quad p_{\text{Fail}} = 0.2$$

So:

$$\text{Gini} = 1 - (0.8)^2 - (0.2)^2 = 1 - 0.64 - 0.04$$

$$\boxed{\text{Gini} = 0.32}$$

Compare:

$$0 < 0.32 < 0.5$$

Interpretation

This node is more pure than a 50–50 node, but not perfectly pure.

Understanding Gini Values

For a binary classification problem:

Pass	Fail	Gini
100%	0%	0.00
90%	10%	0.18
80%	20%	0.32
70%	30%	0.42
50%	50%	0.50

Remember

Lower Gini = Better Purity

Gini Before and After Splitting

Suppose the parent node contains:

6 Pass, 4 Fail

Then:

$$Gini_{Parent} = 1 - \left(\frac{6}{10}\right)^2 - \left(\frac{4}{10}\right)^2 = 1 - 0.36 - 0.16$$

$$Gini_{Parent} = 0.48$$

Now suppose we split the node using:

$$Attendance > 75?$$

We must calculate the impurity of the child nodes.

Child Nodes After Splitting

Suppose the split produces:

Left Child

1 Pass

4 Fail

Total:

$$n_L = 5$$

Right Child

5 Pass

0 Fail

Total:

$$n_R = 5$$

We now calculate:

$$Gini_L \quad \text{and} \quad Gini_R$$

Gini of the Left Child

Left child:

1 Pass, 4 Fail

Therefore:

$$p_{\text{Pass}} = \frac{1}{5} = 0.2, p_{\text{Fail}} = \frac{4}{5} = 0.8$$

Hence:

$$Gini_L = 1 - (0.2)^2 - (0.8)^2 = 1 - 0.04 - 0.64$$

$$Gini_L = 0.32$$

Gini of the Right Child

Right child:

5 Pass, 0 Fail

Therefore:

$$p_{\text{Pass}} = 1$$

$$p_{\text{Fail}} = 0$$

Hence:

$$Gini_R = 1 - (1)^2 - (0)^2$$

$$Gini_R = 0$$

Weighted Gini After Splitting

Because the two child nodes may contain different numbers of observations, we calculate a **weighted Gini**.

$$Gini_{Split} = \frac{n_L}{n} Gini_L + \frac{n_R}{n} Gini_R$$

Here:

$$n_L = 5, \quad n_R = 5, \quad n = 10$$

Therefore:

$$Gini_{Split} = \frac{5}{10}(0.32) + \frac{5}{10}(0)$$

$$Gini_{Split} = 0.16$$

Was the Split Useful?

Before splitting:

$$Gini_{Parent} = 0.48$$

After splitting:

$$Gini_{Split} = 0.16$$

Therefore:

$$0.48 \rightarrow 0.16$$

Conclusion

The impurity decreased substantially.
Therefore, the split is useful.

Choosing Between Two Candidate Splits

Suppose two possible splits produce:

$$Gini_{Split\ A} = 0.16 \text{ and } Gini_{Split\ B} = 0.38$$

Which split should the tree choose?

Choose Split A

because:

$$0.16 < 0.38$$

Rule

Choose the split with the **smallest weighted impurity**.

Entropy: Another Measure of Impurity

Entropy measures uncertainty or disorder.

$$H = - \sum_{k=1}^K p_k \log_2(p_k)$$

For two classes:

$$H = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

Interpretation

- Entropy = 0 $\rightarrow$ pure node
- Higher entropy $\rightarrow$ more uncertainty
- For two equally represented classes, entropy is maximum

Entropy Example 1: Pure Node

Suppose:

10 Pass, 0 Fail

Then:

$$p_{\text{Pass}} = 1, p_{\text{Fail}} = 0$$

So:

$$H = -1 \log_2(1)$$

Since:

$$\log_2(1) = 0$$

we get:

$$H = 0$$

Meaning

There is no uncertainty about the class.

Entropy Example 2: Maximum Uncertainty

Suppose:

5 Pass, 5 Fail

Then:

$$p_{\text{Pass}} = 0.5, p_{\text{Fail}} = 0.5$$

Therefore:

$$H = -0.5 \log_2(0.5) - 0.5 \log_2(0.5)$$

Since:

$$\log_2(0.5) = -1$$

we obtain:

$$H = 1$$

Interpretation

The classes are equally mixed, so uncertainty is maximum.

Entropy Example 3

Suppose:

8 Pass, 2 Fail

Then:

$$p_1 = 0.8, \quad p_2 = 0.2$$

Entropy:

$$H = -0.8 \log_2(0.8) - 0.2 \log_2(0.2)$$

Approximately:

$$H = 0.722$$

Therefore:

$$H \approx 0.72$$

$$0 < 0.72 < 1$$

So the node is partially pure.

Information Gain

Entropy is often used through a measure called:

Information Gain

$$IG = H(\text{Parent}) - H(\text{Children})$$

where:

$$H(\text{Children}) = \sum_j \frac{n_j}{n} H_j$$

Interpretation

Information Gain measures how much uncertainty is reduced after splitting.

Information Gain Example

Suppose: $H(\text{Parent})=0.97$ and after splitting: $H(\text{Children})=0.42$

Then:

$$IG = 0.97 - 0.42$$

$$IG = 0.55$$

If another split has:

$$IG = 0.20$$

then the tree prefers:

$$IG = 0.55$$

Rule

Higher Information Gain means a better split.

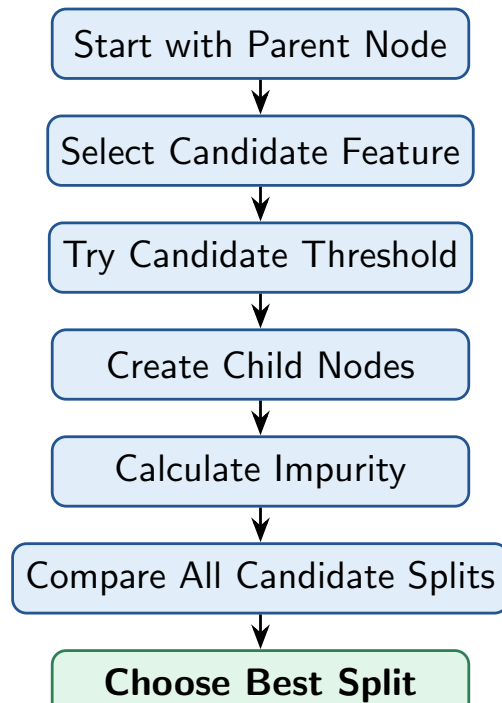
Gini vs Entropy

Property	Gini	Entropy
Measures	Impurity	Uncertainty
Minimum value	0	0
Pure node	0	0
Binary maximum	0.5	1
Good split	Lower Gini	Higher Info. Gain
Computation	Simpler	Uses logarithm
Common use	CART	ID3/C4.5-style ideas

Important

In practice, Gini and entropy often produce similar trees.

Full Split Selection Process



Worked Dataset

Consider:

Attendance	Marks	Result
90	80	Pass
85	70	Pass
80	65	Pass
75	55	Pass
70	50	Fail
65	45	Fail
60	35	Fail
55	30	Fail

Possible first questions: Attendance > 72.5 ? or Marks > 52.5 ?

Class Activity

Calculate the impurity produced by each split and decide which is better.

Candidate Split: Attendance

Try:

$$\text{Attendance} > 72.5$$

This gives:

Left Child

Attendance ≤ 72.5

4 Fail

0 Pass

$$Gini_L = 0$$

Right Child

Attendance > 72.5

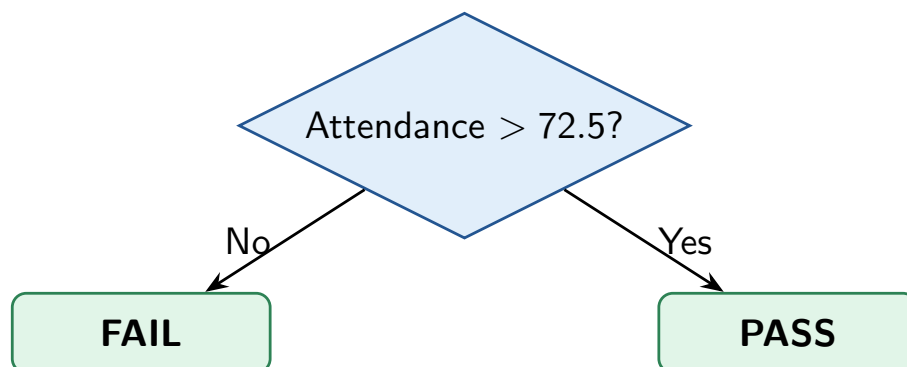
4 Pass

0 Fail

$$Gini_R = 0$$

Therefore: $Gini_{Split} = 0$ Perfect separation!

Resulting Decision Tree



Observation

One single split perfectly separates the training observations.

29. Important Warning

A split that perfectly separates the **training data** is not always perfect for new data.



This leads to an important topic

Overfitting

We will study this in the next section.

Quick Practice: Gini

A node contains:

12 observations

with:

9 Pass and 3 Fail

Calculate:

- 1 p_{Pass}
- 2 p_{Fail}
- 3 Gini impurity

Hint

$$Gini = 1 - p_{\text{Pass}}^2 - p_{\text{Fail}}^2$$

Solution to the Practice Problem

Given:

9 Pass, 3 Fail

Total:

$$n = 12$$

Therefore:

$$p_{\text{Pass}} = \frac{9}{12} = 0.75$$

$$p_{\text{Fail}} = \frac{3}{12} = 0.25$$

Hence:

$$Gini = 1 - (0.75)^2 - (0.25)^2 = 1 - 0.5625 - 0.0625$$

$$Gini = 0.375$$

Quick Practice: Entropy

A node contains:

4 Pass and 4 Fail

Calculate entropy.

$$H = - \sum p_k \log_2(p_k)$$

Think Before Calculating

Are the two classes equally represented?

What should happen to uncertainty?

Solution: Entropy

Here:

$$p_{\text{Pass}} = \frac{4}{8} = 0.5$$

and

$$p_{\text{Fail}} = 0.5$$

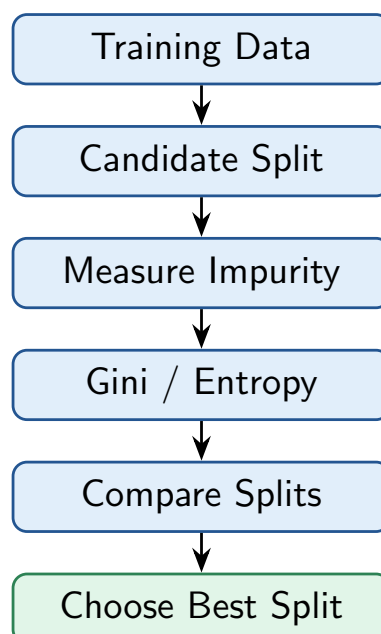
Therefore:

$$H = -0.5 \log_2(0.5) - 0.5 \log_2(0.5)$$

$$= 0.5 + 0.5$$

$$H = 1$$

Section 2 Summary



Key Formulas to Remember

Gini Impurity

$$Gini = 1 - \sum_{k=1}^K p_k^2$$

Entropy

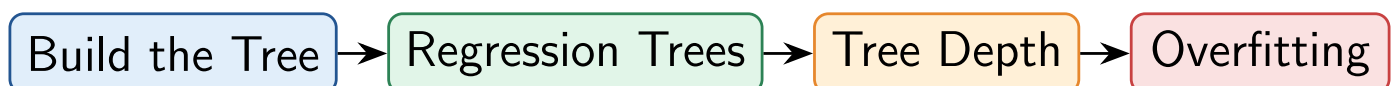
$$H = - \sum_{k=1}^K p_k \log_2(p_k)$$

Information Gain

$$IG = H(\text{Parent}) - H(\text{Children})$$

Coming Next: Section 3

Now We Know How to Choose a Split



Section 3: Building and Controlling Decision Trees